How Label-Efficient is the MNIST Record Holder? A Low-Data Benchmark and a Stacking Pitfall

Attila Torda
Independent Researcher

Abstract

The reproducible MNIST CNN record holder—a majority vote of three "simple" CNNs—is celebrated for accuracy (99.87%), but its *label efficiency* is rarely measured. On a class-balanced low-label sweep ($n = 20$ to 5000, 5 seeds) we find it is already near the data-efficiency frontier: it reaches 64% from 20 labels and 90% from 100, and **wins at every budget** against a strong, diverse ensemble of an interpretable skeleton-graph classifier, a diffusion-augmented CNN, and the record holder's own networks. The ensemble converges to within $\sim$2 pp by $n \geq 1000$ but never overtakes. We then test whether *augmenting* the record holder with this structural and generative diversity helps, and surface a methodological pitfall: a leakage-free stacker that reserves a held-out split to fit its meta-learner *starves the base learners of scarce labels*, inflating the apparent ensemble gain by $\sim$3 pp at $n = 100$–200. Measured against a baseline that keeps all its labels, the gain shrinks to $+0.7$ pp or reverses. We release a reproducible benchmark and a faithful light re-implementation of the record holder, and argue that low-data ensembling results must be reported against full-budget baselines.

1 Introduction

Strong CNNs are near-perfect on MNIST [5]; the reproducible record is a majority vote of three deep "simple" CNNs with kernels $3, 5, 7$, reaching 99.87% [1]. Two questions are less studied. First, *how label-efficient* is such a model—how far does it fall at 20, 100, or 1000 labels, and can a diverse ensemble of cheaper, decorrelated learners beat it there? Second, if a single strong model cannot be beaten head-to-head, can it be *augmented* by adding structural and generative members? We benchmark both questions on a class-balanced low-label sweep, contribute a faithful light re-implementation of the record holder, and report an honest negative with a useful methodological lesson about low-data stacking.

2 Setup

Models. The **record holder** [1] is re-implemented faithfully but lightly (one model per kernel, fewer epochs); our repro reaches 99.65% at 60k (vs published 99.87%). The **grand ensemble** stacks three decorrelated members: a skeleton-fusion CNN, a 93-dim skeleton-graph bag-of-features with a random forest [2, 3], and a CNN trained on images from a small class-conditional diffusion model [4, 6]; members are combined by leakage-free stacked generalisation [7].

Protocol. For each budget $n \in \{20, \ldots, 5000\}$ we sample exactly $n/10$ labels per digit, train on the same budget, and evaluate on the full 10k test set, averaged over 5 seeds. The diffusion member is active only for $n \geq 500$ (a DDPM cannot be trained on fewer images).

38th Conference on Neural Information Processing Systems (NeurIPS 2024).

Table 1: MNIST test accuracy (%, mean over 5 seeds). The record holder wins at every budget.

n	record holder	grand ensemble	grand ensemble + aug
20	**64.2**	44.1	46.7
50	**83.4**	58.4	61.3
100	**90.2**	80.9	81.1
200	**96.2**	86.6	89.6
500	**98.2**	92.6	95.5
1000	**98.8**	95.0	96.9
2000	**99.1**	96.7	98.0
5000	**99.4**	97.8	98.7
60000	**99.65**	—	—

Table 2: The held-out-split pitfall. "Apparent" compares the augmented model to a stacker baseline trained on 70% of labels; "true" compares it to a record holder trained on all n labels. The apparent low-data gain is largely an artefact.

n	augmented	baseline (70%)	apparent Δ	record holder (full n)	true Δ
100	90.9	86.7	$+4.2$	90.2	$+0.7$
200	95.4	93.2	$+2.1$	96.2	-0.9
500	97.6	97.6	$+0.1$	98.2	-0.6

3 Results

The record holder is the data-efficiency winner. Table 1 and Figure 1 show MNIST test accuracy across budgets. The record holder wins at *every* budget—most decisively in the extreme low-data regime (64% vs 47% at $n=20$), exactly where explicit structural priors were expected to help most. The grand ensemble (with morphological augmentation) closes the gap monotonically, reaching within $\sim$2 pp by $n \geq 1000$, but never overtakes. Simple rotation/translation augmentation on a deep CNN is more label-efficient than hand-crafted skeleton/structural features.

Augmenting the record holder, and a stacking pitfall. If the ensemble cannot win head-to-head, can its members *augment* the record holder? We stack the record holder's M3/M5/M7 with the structural and diffusion members (Figure 1). A leakage-free stacker fits its meta-learner on a held-out 30% of the budget, so the base learners see only 70%. Against *that* (weakened) record-holder baseline the augmented model appears to gain $+4.2$ pp at $n=100$ and $+2.1$ at $n=200$. But the comparison is unfair: the baseline was starved of labels. Measured against a record holder that keeps *all* n labels (Table 2), the gain shrinks to $+0.7$ pp at $n = 100$ and *reverses* to -0.9 at $n = 200$. Most of the apparent ensemble gain was the held-out split silently recovering, not new signal.

4 Discussion

Two lessons. (1) **SOTA CNNs are already near the label-efficiency frontier on MNIST**: a diverse ensemble of structural and generative learners—each individually reasonable—does not beat a well-tuned deep CNN ensemble at any budget down to $n = 20$. Hand-crafted topology helps less than geometric augmentation even when labels are scarcest. (2) **Low-data ensembling results must use full-budget baselines**: leakage-free stacking that reserves a held-out split for the meta-learner systematically disadvantages the baseline by removing scarce labels, and can manufacture a multi-point "gain" that is really label recovery. Held-out-free combiners (soft or confidence-weighted averaging) avoid this and let every learner use all n labels.

Limitations. A single dataset (MNIST) and a light record-holder repro (99.65% vs 99.87%); 5 seeds; the diffusion member needs $n \geq 500$. The negative result is specific to clean MNIST—under *corruption*, corruption-agnostic structural/generative diversity does help the record holder, which we study separately.

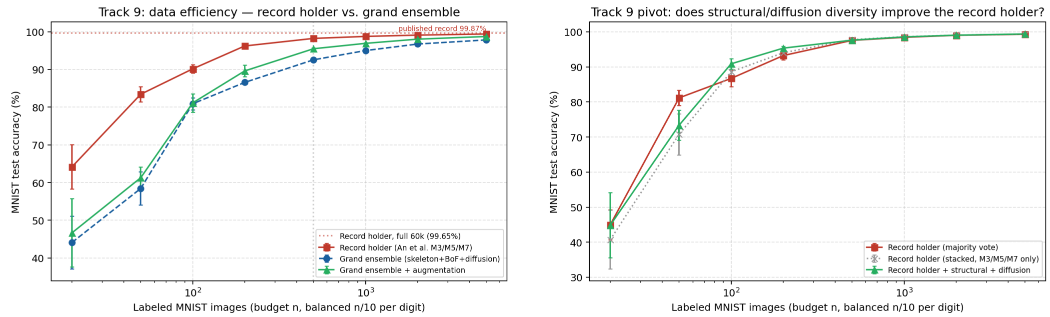

Figure 1: Left (1): data-efficiency—the record holder (red) leads the grand ensemble at every budget. Right (1): augmenting the record holder; the apparent low-data gain over the 70%-trained stacker baseline largely vanishes against a full-budget baseline.

5 Conclusion

The MNIST record holder is remarkably label-efficient and is not beaten, nor cleanly augmented, by structural and generative diversity in the clean low-data regime. The apparent augmentation gain is largely a held-out-split artefact. We release the benchmark and the record-holder repro, and recommend full-budget baselines for all low-data ensembling claims.

Acknowledgments and Disclosure of Funding

Reproducibility: `scripts/run_track9_benchmark.py` (data efficiency), `scripts/run_track9_augment.py` (augmentation + pitfall). Models: `src/track9/`. Raw results: `track9_results.json`, `track9_augment_results.json`.

References

[1] Sanghyeon An, Minjun Lee, Sanglee Park, Heerin Yang, and Jungmin So. An ensemble of simple convolutional neural network models for MNIST digit recognition. *arXiv preprint arXiv:2008.10400*, 2020.

[2] Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001.

[3] Zicheng Guo and Richard W. Hall. Parallel thinning with two-subiteration algorithms. *Communications of the ACM*, 32(3):359–373, 1989.

[4] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

[5] Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998.

[6] Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. In *International Conference on Learning Representations (ICLR)*, 2021.

[7] David H. Wolpert. Stacked generalization. *Neural Networks*, 5(2):241–259, 1992.